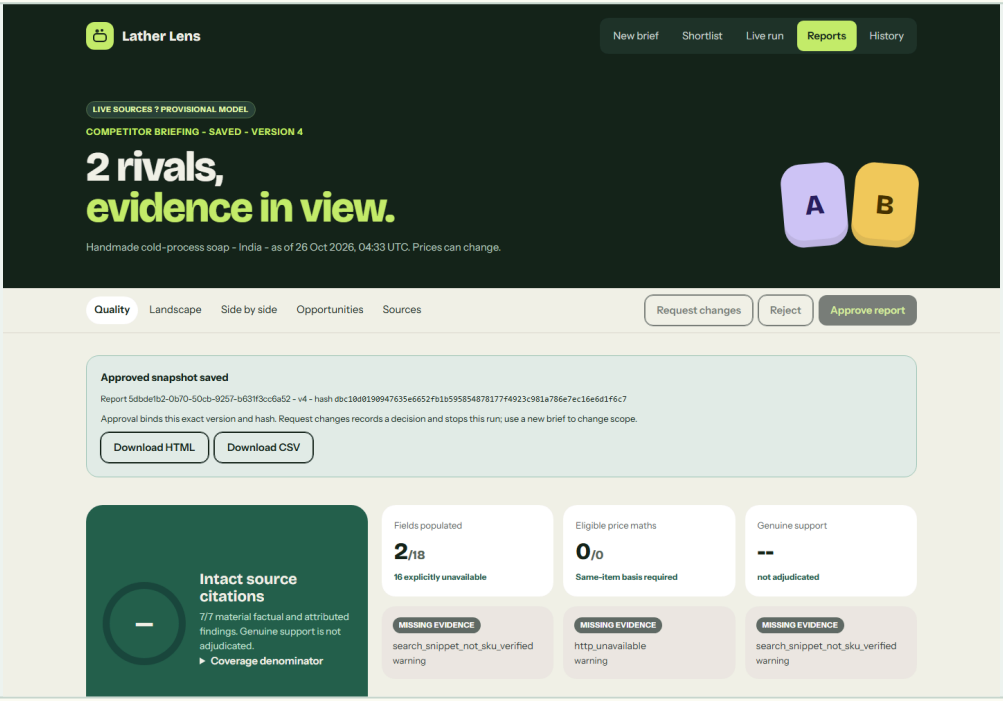
PROJECT 3A / MARKET RESEARCH AGENT

Lather Lens

Evidence-backed competitor research for a handmade-soap founder.

LIVE PILOT - complete - report v4



The one-line goal

Help a founder discover relevant competitors, gather public evidence and review a structured comparison in one web application, with human approval at shortlist and final report.

26 October 2026. Actual local application screenshots. A saved live run is dated evidence, not a fresh search performed during the recording.

Reasoning inside durable boundaries.

Brief → Scout → Human shortlist review

↓

Parallel Researchers → Normalize → Analyst

↓

Independent Reviewer → Bounded correction

↓

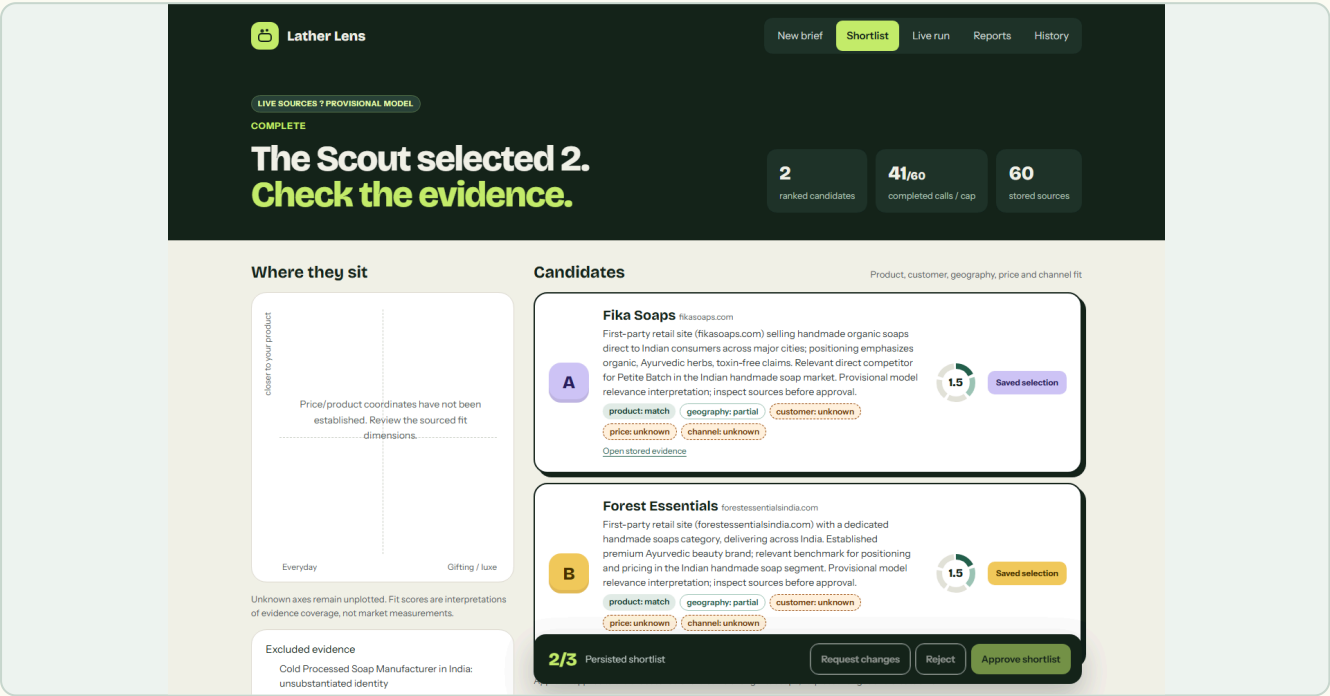
Human report approval → Immutable save/export

Role	Responsibility
Scout	Tavily discovery, model-assisted relevance, source identity validation; up to three competitors.
Researcher	Public web/news and first-party product evidence; separate competitor lanes.
Analyst	Source-attributed comparison and conditional opportunities. Deterministic arithmetic stays in code.
Reviewer	Independent evidence/arithmetic checks and a separate model critique; unresolved issues block approval.
Orchestrator	LangGraph state, checkpoints, retries and human interrupts. Coordination is deterministic.

Stack and model boundary

Next.js, TypeScript and Tailwind; FastAPI/Pydantic; SQLAlchemy/Alembic; Supabase Auth/Postgres with owner RLS. A provider-neutral model port isolates the provisional Fireworks configuration. LangSmith remote export is off.

A shortlist you can inspect.



Actual pilot scope

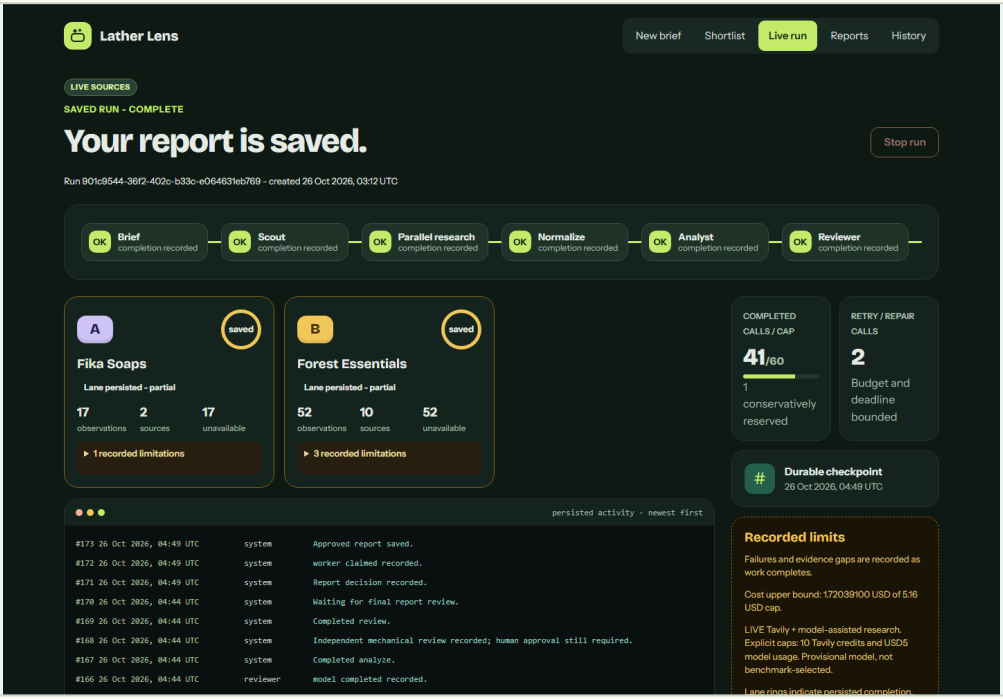
Petite Batch; handmade cold-process soap; India. Fika Soaps and Forest Essentials passed discovery checks and were explicitly approved by the user. The system retained the reason for fewer than three candidates.

Use the local application

1. Open the signed-in project browser with `node scripts/open-local.cjs --live --report`.
2. Inspect Shortlist, Live Run and Report for this same run.
3. Review the AI briefing, limitations and clickable evidence.
4. Approve only a report you accept; then download HTML/CSV after saved state.

The local session helper uses the existing disposable local identity. Keys and tokens are never included in the submission. Hosting is not required by the supplied handout.

Show the work. Keep the unknowns.



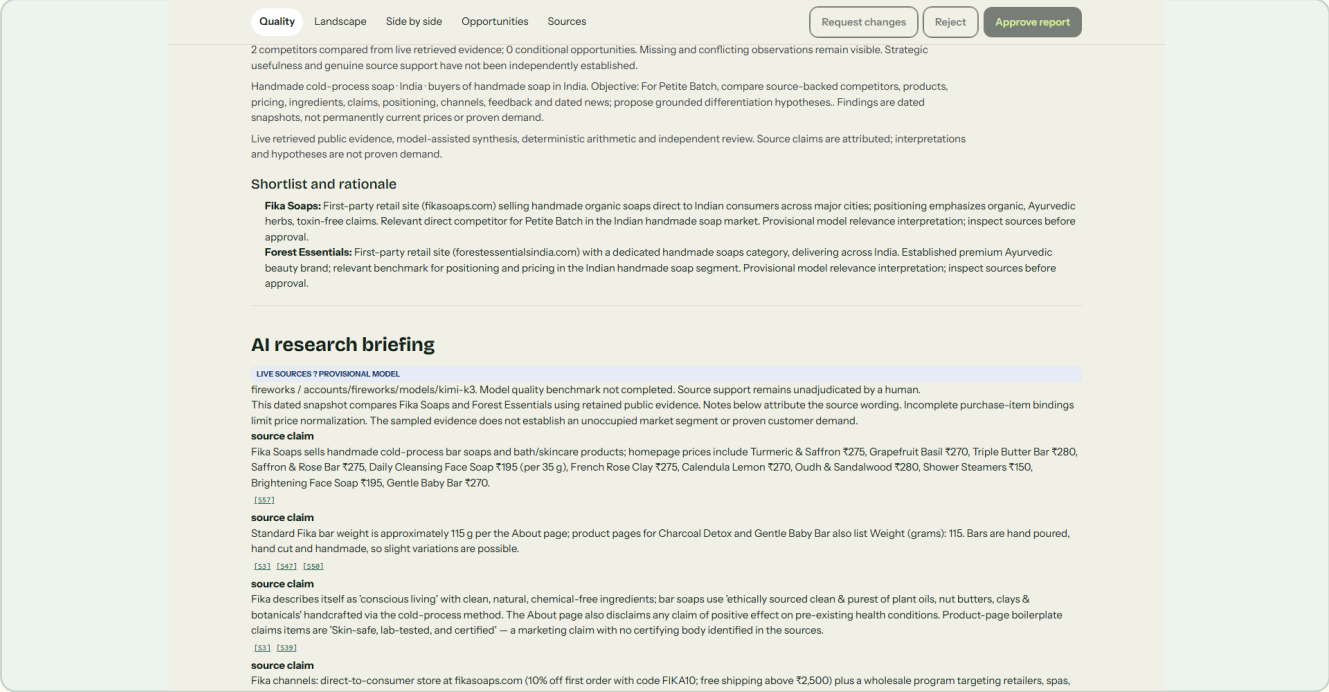
Postgres retains retrieved URLs, times, excerpts and content hashes, durable events and checkpoints. The pilot recovered from an invalid credential, malformed JSON, missing evidence and a citation validation failure without rerunning completed research.

What this run measured

Model tokens recorded	106461
Reconciled total cost upper bound	USD 1.72039100
Known deterministic observations	7
Final approval state	saved
Independent model critique	ready_for_human

Cost is a configured-rate estimate plus conservative unknown-attempt reservations, not a provider invoice. No throughput or production scalability claim is made.

Claims need more than a citation.



A valid URL and intact quote establish traceability, not entailment. The independent model reviewer can flag wrong prices, dates and unsupported wording, but genuine support still requires claim-level adjudication. Unknown SKU/pack bindings prevent price-per-100g arithmetic.

Acceptance and limitations

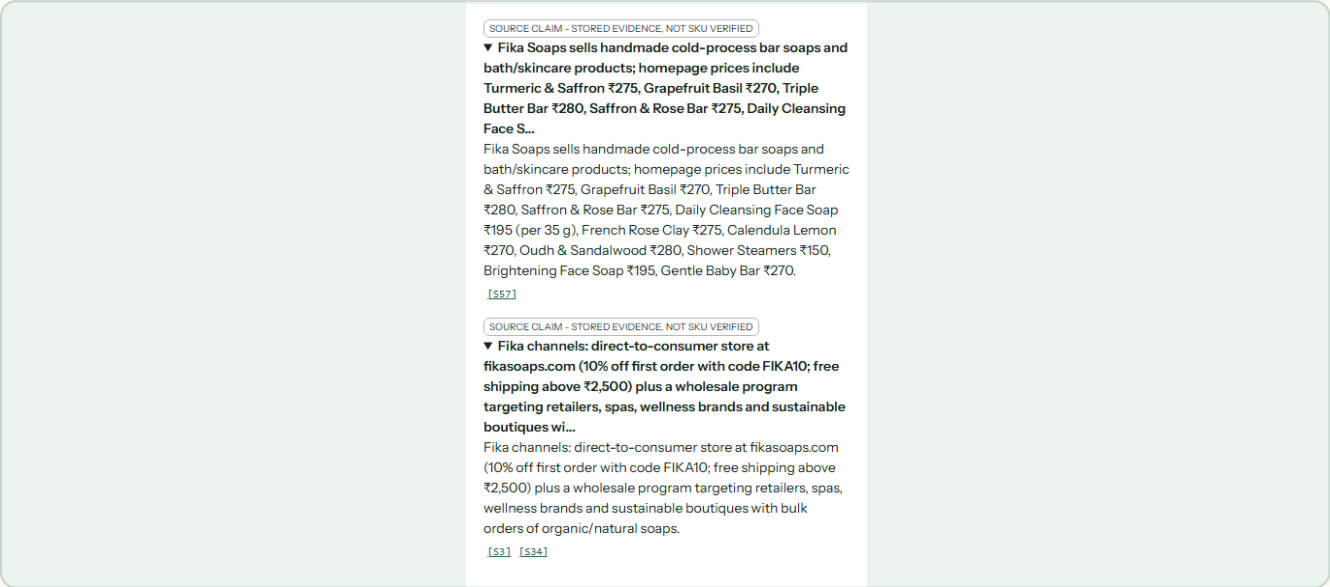
- Original BRD: at least 90% material-claim traceability; at least 4/5 owner-rated useful reports; at least 4/5 failure tests handled; resume without losing completed work; no knowingly fabricated facts.
- Owner usefulness and full real-claim support remain unmeasured. Offline test passes are separate from business acceptance.
- Model benchmark design exists; final model selection is not complete.
- Website restrictions, missing SKU/weights and stale announcements remain visible. No external publishing or deployment has occurred.

Submission: clean source repository/ZIP, this illustrated overview, a 3.5-minute walkthrough and an approved sourced sample when available. Preserve historical documents locally; exclude secrets, caches and account screenshots.

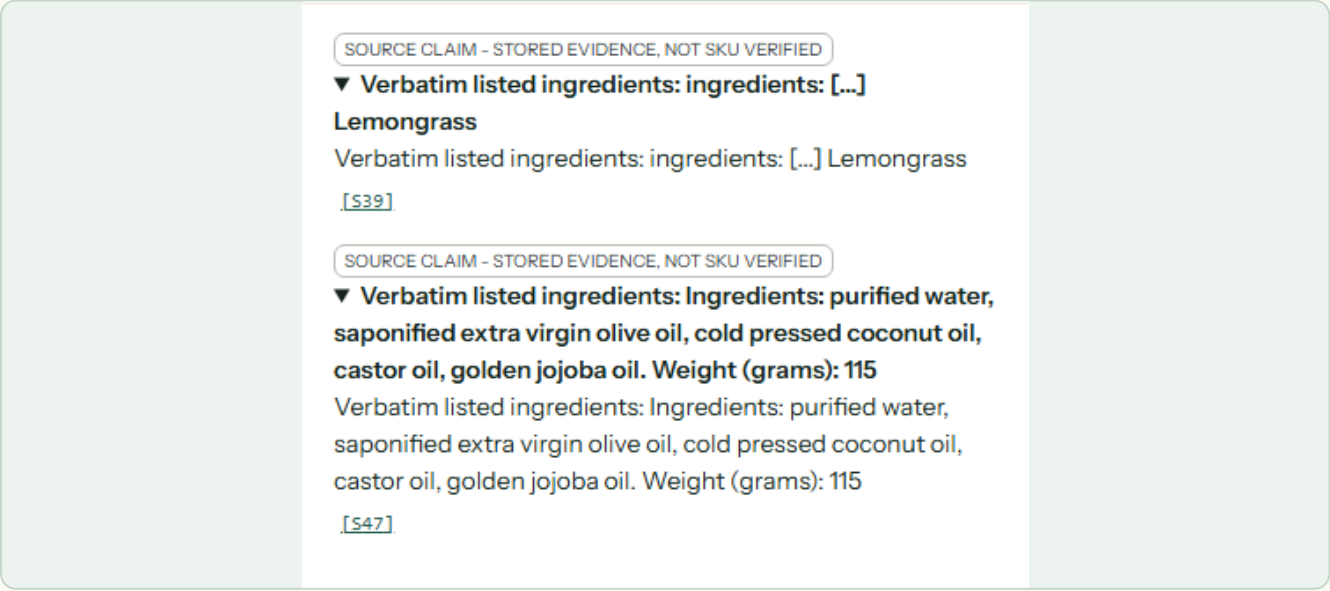
The results were in the briefing.

The comparison reuses persisted source-claimed notes and explicit declarations in stored evidence when structured fields are missing. These remain attributed claims; they do not create a verified SKU or change the saved report hash.

Listed-price evidence

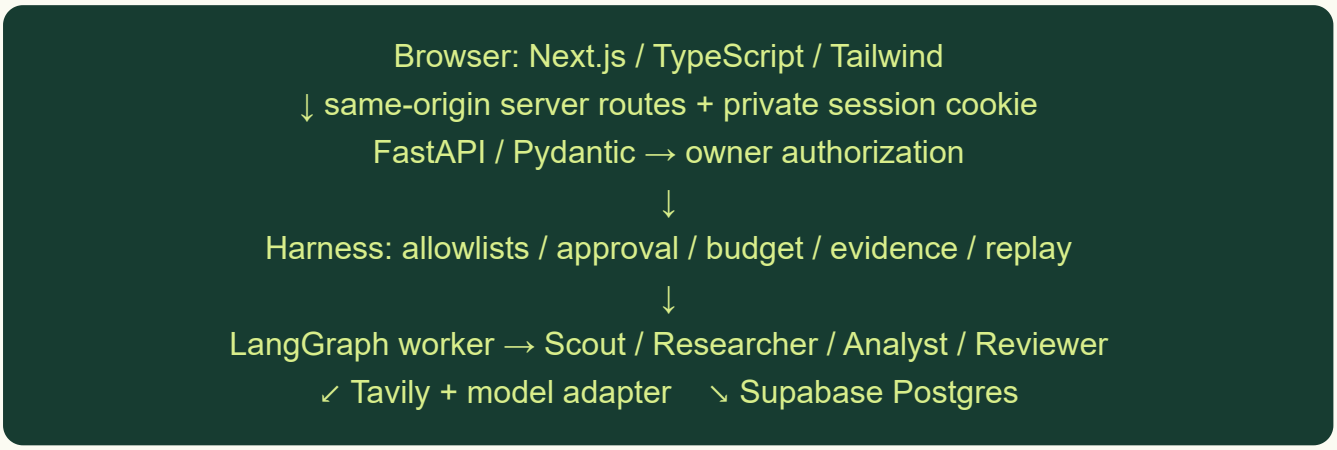


Ingredient evidence



Actual comparison-cell screenshots, enlarged for readability. Source buttons open stored evidence. All cited sources must belong to this competitor; missing, unavailable and mixed-brand bindings are excluded.

How the application fits together.



Layer	What we used and why
Interface	Next.js App Router, React, TypeScript, Tailwind CSS; Brief, Shortlist, Live Run, Report and History.
Backend	Python FastAPI and Pydantic typed requests/results; SQLAlchemy repositories and Alembic migrations.
State and identity	Supabase Auth and Postgres; owner RLS, durable checkpoints, fenced worker lease and heartbeat, immutable report versions.
Search API	Tavily Search API through the harness only; public HTTPS retrieval validates destinations, redirects and access restrictions.
Model API	Provider-neutral ModelAdapter port. The authorized pilot uses Fireworks JSON chat with Kimi K3 provisionally. Keys stay server-side; no final benchmark winner.
Observability	Persisted event journal and sanitized LangSmith integration. Remote LangSmith export is disabled.
Validation	pytest, Ruff, mypy; Vitest, ESLint, TypeScript; Playwright screenshots and 17 deterministic golden cases.

Application API examples

`POST /api/briefs` saves a brief; `GET /api/workspace` reads persisted results; shortlist/report decision routes record version-bound approval; the run export route returns approved HTML/CSV. FastAPI publishes OpenAPI schemas. No API key is exposed to the browser.

This submission runs locally. No cloud deployment, hosted production capacity or final business acceptance pass is claimed.